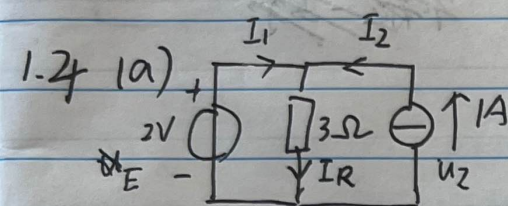


第一次作业

$$1.1 \quad U_A = \frac{P}{I_A} = 5V$$

$$I_B = \frac{P}{U_B} = -1A$$

$$U_C = \frac{P}{I_C} = -10V$$



该电阻上通过电流为 I_R

$$\text{由 KVL, } I_R \cdot 3\Omega = 2V \Rightarrow I_R = \frac{2}{3}A$$

$$\therefore I_2 = 1A \quad \text{由 KCL, } I_1 = -\frac{1}{3}A$$

$$\text{由 KVL, } U_2 = I_R \cdot 3\Omega = 2V$$

$$\therefore P_R = U_R I_R = I_R^2 R = \frac{4}{3}W$$

$$\text{电压源 } P_u = U_1 I_1 = -\frac{2}{3}W$$

$$\text{电流源 } P_i = U_2 I_2 = 2W$$

1b) 由 KVL, 设电源上电压为 U_1

$$\text{有 } 2V + I_R \cdot 3\Omega = U_1$$

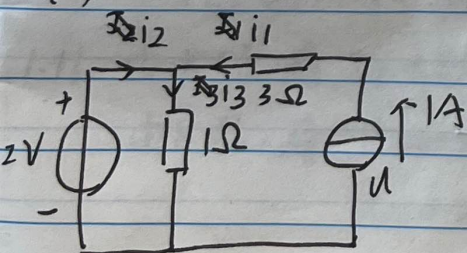
$$\Rightarrow U_1 = 5V$$

$$\therefore P_R = U_R I_R = 3W$$

$$\text{电压源 } P_u = U_1 I_1 = -2W$$

$$\text{电流源 } P_i = U_1 I_2 = 5W$$

1c)



$$\text{由 KVL, 有 } 2V = I_3 \cdot 1\Omega \Rightarrow I_3 = 2A$$

$$\therefore I_1 = 1A \quad \text{由 KCL, } I_2 = 1A$$

$$\text{由 KVL, 有 } U = I_1 \cdot 3\Omega + I_3 \cdot 1\Omega$$

$$\Rightarrow U = 5V$$

$$\therefore 3\Omega \text{ 电阻 } P_1 = I_1^2 R = 3W$$

$$1\Omega \text{ 电阻 } P_2 = I_3^2 R = 4W$$

$$\text{电压源 } P_u = U_1 I_1 = 2W \quad \text{电流源 } P_i = U_1 I_2 = 5W$$

由KCL

1.6 节点①有 $i_2 = i_3 + 1A$

节点②有 $1A = i_1 + 2A$

节点③有 $2A + 3A = i_4$

节点④有 $i_3 + i_4 = 8A$

~~节点⑤有 $i_1 + 8A = i_2 + 3A$~~

$i_1 = -1A$

$\Rightarrow i_2 = 4A$

$i_3 = 3A$

$i_4 = 5A$

~~不能~~

~~1.7~~ 对线 ① $\rightarrow$ ④ $\rightarrow$ ③

有 $i_2 + 2A + 3A = 8A + 1A \Rightarrow i_2 = 4A$

1.8 由KCL

1.7 1) ①节点有 $i_1 + 2A + 1A = 0$

$i_1 = -3A$

②节点有 $i_1 + 1A = i_4$

$\Rightarrow i_4 = -2A$

③节点有 $i_4 + 1A = i_3$

$i_3 = -1A$

④节点有 $i_2 + i_3 + 1A = 0$

$i_2 = 0$

不能，只有5个节点，可列4个方程，允许4个未知量

2) ~~①④⑤网孔有 $U_{14} = 7V$~~ $U_{14} = U_{15} + 7V$ $U_{15} - 7V - U_{15} = 0$

~~②③⑤网孔有 $U_{52} + 6V = 0$~~ $U_{52} + 6V - U_{53} = 0$

~~①②⑤网孔有 $U_{15} = 8V$~~ $U_{15} + U_{52} - 5V = 0$

~~⑤③④网孔有 $U_{53} = 7V$~~ $7V - 8V - U_{53} = 0$

$\Rightarrow U_{53} = -1V$

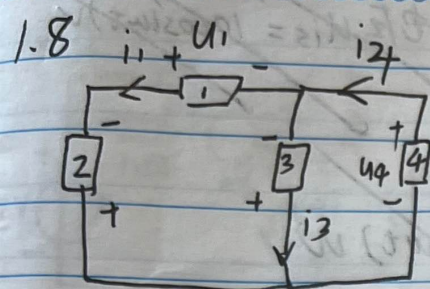
$U_{52} = -7V$

$U_{15} = 12V$

$U_{14} = 19V$

不能，只能列出4个独立方程

求4个未知量



由KCL, 有 $i_2 - i_3 - i_1 = 0$

$$\Rightarrow i_2 = -1A$$

由KVL, 有 $u_1 + u_2 - u_3 = 0$

$$u_4 + u_3 = 0$$

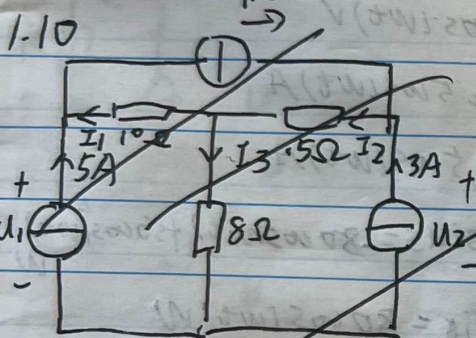
$$\Rightarrow u_3 = 5V \quad u_2 = -5V$$

$$\therefore P_1 = u_1 i_1 = -20W$$

$$P_2 = u_2 i_1 = 10W$$

$$P_3 = u_3 i_3 = 15W$$

$$P_4 = u_4 i_4 = -5W$$



由KCL, $i_1 + 5A = 7A$

$$i_1 = 2A$$

$$7A + 3A = i_2$$

$$i_2 = 10A$$

$$i_2 = i_1 + i_3$$

$$i_3 = 8A$$

由KVL, 有 $u_1 + i_1 \cdot 10\Omega - i_3 \cdot 8\Omega = 0$

$$u_2 - i_2 \cdot 5\Omega - i_3 \cdot 8\Omega = 0$$

$$\Rightarrow u_1 = 44V$$

$$u_2 = 114V$$

$$\therefore P_1 = u_1 i_1 = 220W$$

$$P_2 = u_2 \cdot 3A = 342W$$

1.11 由KCL, 有 $I_1 + I_2 - I_7 = 0 \Rightarrow I_1 = 1A$

由KVL, 有 $u_{24} + u_{43} + u_{32} = 0$

$$u_{13} + u_{32} + u_{21} = 0$$

$$\Rightarrow u_{21} = 6V \quad \therefore P = -u_{21} \cdot I_1 = -6W$$

有 $i_1 + 0.3A = i$ (KCL)

由KVL, $30V = i \cdot 60\Omega$

$$u + 60\Omega \cdot i_1 - 0.3A \cdot 50\Omega = 0$$

$$\Rightarrow i_1 = 0.5A \quad i = 0.8A$$

$$u = -15V \quad P_u = u i = 24W$$

$$P_i = u i = -4.5W$$

1.12 (a) 由KVL知 电阻上电压 $U_R = 10 \cos(10t) \text{ V}$ 电压源上电压 $U_{is} = 10 \cos(10t) \text{ V}$

$$I_R = i_R = \frac{U_R}{R} = 5 \cos(10t) \text{ A}$$

$$\therefore i_{is} = -3 \cos(10t) \text{ A}$$

$$P_u = -U_{is} i_{is} = -\frac{5 \cdot 10}{\sqrt{2} \cdot \sqrt{2}} = -15 \text{ W} \quad -30 \cos^2(10t) \text{ W}$$

$$P_i = i_{is} U_{is} = \frac{8}{\sqrt{2}} \cdot \frac{10}{\sqrt{2}} = 40 \text{ W} \quad 80 \cos^2(10t) \text{ W}$$

$$P_R = U_R i_R = \frac{10}{\sqrt{2}} \cdot \frac{5}{\sqrt{2}} = 25 \text{ W} \quad 50 \cos^2(10t) \text{ W}$$

1b) 电阻上电压 $U_R = i_R R = 16 \cos(10t) \text{ V}$

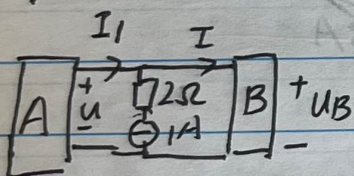
由KVL, $U_{is} = U_s + U_R = 26 \cos(10t) \text{ V}$

$$\therefore P_u = -U_{is} i_{is} = -40 \text{ W} \quad -80 \cos^2(10t) \text{ W}$$

$$P_i = U_{is} i_{is} = 104 \text{ W} \quad 208 \cos^2(10t) \text{ W}$$

$$P_R = U_R i_R = 64 \text{ W} \quad 128 \cos^2(10t) \text{ W}$$

1.17



由KCL, $I_1 + 1\text{A} = I$

$$\Rightarrow I_1 = 1\text{A}$$

由KVL, 有 $U + U_R = U_{is}$

$$\Rightarrow U_{is} = 12\text{V}$$

$$\text{有 } U_B + -U_{is} + U_R = 0$$

$$\Rightarrow U_B = 10\text{V}$$

$$\therefore P_i = U_{is} i_{is} = 12\text{W}$$

$$P_A = U I_1 = 10\text{W}$$

$$P_B = -U_B \cdot I = -20\text{W}$$

(a) 1.12. 由KVL知电阻上电压

$$U_R = 10 \cos(10t) \text{ V}$$

$$U_{is} = 10 \cos(10t) \text{ V}$$

$$i_R = \frac{U_R}{R} = 5 \cos(10t) \text{ A}$$

$$i_{is} = 8 - 5 \cos(10t) \text{ A}$$

$$P_u = -U_{is} i_{is} = -80 \cos(10t) + 50 \cos^2(10t) \text{ W}$$

$$P_i = i_{is} U_{is} = 80 \cos(10t) \text{ W}$$

$$P_R = U_R i_R = 50 \cos^2(10t) \text{ W}$$

1b) 电阻上电压 $U_R = i_R R = 16 \cos(10t) \text{ V}$

由KVL, $U_{is} = U_s + U_R$

$$= 10 + 16 \cos(10t) \text{ V}$$

$$P_u = -U_{is} i_{is} = -80 \cos(10t) \text{ W}$$

$$P_i = U_{is} i_{is} = 80 \cos(10t)$$

$$+ 128 \cos^2(10t) \text{ W}$$

$$P_R = U_R i_R = 128 \cos^2(10t) \text{ W}$$

由KVL, 电压源两端电压 $u=4V$
 由KCL, $i_1 + 0 = 2i + i \Rightarrow i_1 = 3i$ 有 $4\Omega \cdot i + 2V$
~~有 $4\Omega \cdot i + 2V = 1\Omega \cdot i \Rightarrow$~~

由KVL, 有 $4V - 2i = 0$, 显然 $i_2 = 0$

$\Rightarrow i = 2A$ (由K)

由KVL, 有

由KVL, 有 $2V + 4\Omega \cdot i - i_1 \cdot 1\Omega = 0$

$\Rightarrow i_1 = 10A$ 电

$\therefore i_3 = i + i_1 = 12A$

$\therefore P = i_3 \cdot u = 12 \cdot 4 = 48W$

由KVL, 电压源两端电压

$u = 4V$

由KVL, 有 $4\Omega \cdot i + 4V = -2V$

$\Rightarrow i = -0.5A$

$\therefore P = u \cdot i = -4W$

由KCL, $i_2 = i + \beta i$

$i_2 + \beta i = i \therefore i_2 = (1-\beta)i$

由KVL, $u - i_2 R - u_s = 0$

$\Rightarrow u = (1-\beta)iR + u_s$

由KCL, $i_o = i + i_s$

由KVL, 有 $u - (i_o) \cdot R - r i_o = 0$

$\Rightarrow u = (1+i_s)R + r i_o$

~~$= (R+r)i + r i_s$~~

$= (1+i_s)(R+r)$

由KCL, $i_2 = i + g u_2$

有 $u_2 = i_2 R_2$

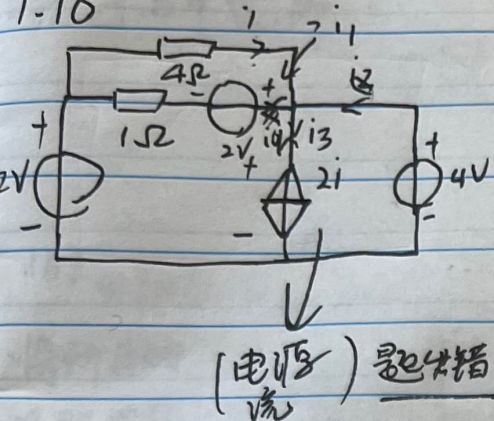
$\Rightarrow i_2 = i + g i_2 R_2$

$\Rightarrow u = (R_1 + r + \frac{R_2}{1-gR_2})i$

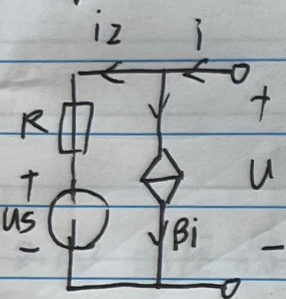
$i_2 = \frac{i}{(1-gR_2)}$

由KVL, $u - i R_1 - r i - i_2 R_2 = 0$

1.18



1.19 (a)



1b)

